

Homework 8

Nov 5, 2025

1 (30'). Let

$$L = \{(\alpha, \beta) : \text{Turing machines } M_\alpha, M_\beta \text{ satisfy } L(M_\alpha)^* = L(M_\beta)\}.$$

Prove that L is undecidable.

2 (40'). Let

$$L = \{\alpha : L(M_\alpha) \text{ contains at least two prime numbers}\},$$

where each input $x \in \{0, 1\}^*$ to the machine M_α is interpreted as a binary number with the most significant bit first. Prove that L is undecidable. (Do not use Rice's theorem.)

3 (30'). Let

$$L = \{(\alpha, x, k) : M_\alpha \text{ halts on input } x \text{ within } k \text{ steps}\},$$

where k is given as a binary number with the most significant bit first. Prove that L is decidable.